Chomsky Normal Form

Example1
Convert the following CFG to Chomsky Normal Form (CNF):

S → aX | Yb
X → S | /\
Y → bY | b

Solution 1
Step 1 - Kill all /\ productions:

By inspection, the only nullable nonterminal is X.
Delete all /\ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without /\ productions, is:
S → aX | a | Yb
X → S
Y → bY | b

Step 2 - Kill all unit productions:

The only unit production is X → S, where the S can be replaced with all S's non-unit productions (i.e. aX, a, and Yb).

The new CFG, without unit productions, is:
S → aX | a | Yb
X → aX | a | Yb
Y → bY | b

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

S → AX | YB | a
X → AX | YB | a
Y → BY | b
A → a
B → b

Step 4 - Shorten the strings of nonterminals to length 2.

All nonterminal strings on the RHS in the above CFG are already the required length, so the CFG is in CNF.

Example2
Convert the following CFG to Chomsky Normal Form (CNF):
S → AA
A → B | BB
B → abB | b | bb

Solution 2

Step 1 - Kill all /\ productions:

There are no /\ productions, so none of the non-terminals is nullable. The CFG remains unchanged.

Step 2 - Kill all unit productions:

The only unit production is A → B, where the B can be replaced with all B's non-unit productions (i.e. all of them).

The new CFG, without unit productions, is:

S → AA
A → BB | abB | b | bb
B → abB | b | bb

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

S → AA
A → BB | XYB | b | YY
B → XYB | b | YY
X → a
Y → b

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

S → AA
A → BB | RB | b | YY
B → RB | b | YY
R → XY
X → a
Y → b

Example3

Convert the following CFG to Chomsky Normal Form (CNF):

S → XaX | bX | Y
X → XaX | XbX | /\
Y → ab

Solution 3

Step 1 - Kill all /\ productions:

By inspection, the only nullable nonterminal is X.
Delete all /\ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without /\ productions, is:

S → XaX | aX | Xa | a | bX | b | Y
X → XaX | aX | Xa | a | XbX | bX | Xb | b
Y → ab

Step 2 - Kill all unit productions:

The only unit production is S → Y, where the Y can be replaced with all Y's non-unit productions (i.e. ab). Furthermore, the Y-production can be completely removed, since its only purpose is to turn S into ab in one particular production sequence: S → Y → ab.

The new CFG, without unit productions, is:

S → XaX | aX | Xa | a | bX | b | ab
X → XaX | aX | Xa | a | XbX | bX | Xb | b

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

S → XAX | AX | XA | BX | AB | a | b
X → XAX | AX | XA | XBX | BX | XB | a | b
A → a
B → b

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

S → XR | AX | XA | BX | AB | a | b
R → AX
X → XR | AX | XA | XQ | BX | XB | a | b
Q → BX
A → a
B → b

Example4

Convert the following CFG to Chomsky Normal Form (CNF):

S → SS | A | B
A → SS | AS | a
B → /\

Solution 4

Step 1 - Kill all /\ productions:

By inspection, the nullable nonterminals are S, A, and B.
Delete all /\ productions and add new productions, with all possible combinations of the nullable nonterminals removed.

The new CFG, without /\ productions, is:

S → SS | A | S
A → SS | AS | S | A | a

Step 2 - Kill all unit productions:

The unit productions S → S and A → A are redundant and can be removed. The only unit production for S is then S → A, which can be replaced with all A's non-unit productions. The only unit production for A is then A → S, which can be replaced with all S's non-unit productions.

The new CFG, without unit productions, is:

S → SS | AS | a
A → SS | AS | a

Step 3 - Replace all mixed strings with solid nonterminals.
Step 4 - Shorten the strings of nonterminals to length 2.

Steps 3 and 4 are not necessary, since the CFG is already in CNF.
The CFG can however be shortened to:
S → SS | a

Example5

Convert the following CFG to Chomsky Normal Form (CNF):

S → XaX | YY | XY
X → /\ | b
Y → Xa

Solution 5

Step 1 - Kill all /\ productions:

By inspection, the only nullable nonterminal is X.
Delete all /\ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without /\ productions, is:
S → XaX | aX | Xa | a | YY | XY | Y
X → b
Y → Xa | a

Step 2 - Kill all unit productions:

The only unit production is S → Y, where the Y can be replaced with all Y's non-unit productions (i.e. all of them).

The new CFG, without unit productions, is:

S → XaX | aX | **Xa** | **a** | YY | XY
X → b
Y → Xa | a

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

S → XAX | AX | XA | YY | XY | a
X → b
Y → XA | a
A → a
B → b

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

S → XR | AX | XA | YY | XY | a
R → AX
X → b
Y → XA | a
A → a
B → b

Example6
Convert the following CFG to Chomsky Normal Form (CNF):

S → aX | bS | ab | X
X → aX | a | /\

Solution 6
Step 1 - Kill all /\ productions:

Both S and X are nullable nonterminals.
Delete all /\ productions and add new productions, with all possible combinations of the nullable S and X removed.

The new CFG, without /\ productions, is:

S → aX | bS | ab | X | a | b
X → aX | a

Step 2 - Kill all unit productions:

The only unit production is S → X, where the X can be replaced with all X's non-unit productions (i.e. aX and a).

The new CFG, without unit productions, is:

S → aX | bS | ab | a | b
X → aX | a

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:
S → AX | BS | AB | a | b
X → AX | a
A → a
B → b

Step 4 - Shorten the strings of nonterminals to length 2.

All nonterminal strings on the RHS in the above CFG are already the required length, so the CFG is in CNF.

Quesiton 7
Convert the following CFG to Chomsky Normal Form (CNF):

S → aX | Y | bab
X → /\ | Y
Y → bb | bXb

Solution 7
Step 1 - Kill all /\ productions:

By inspection, the nullable nonterminal is X.
Delete all /\ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without /\ productions, is:
S → aX | a | Y | bab
X → Y
Y → bb | bXb

Step 2 - Kill all unit productions:

The only unit productions are S → Y and X → Y, where the Y in both cases can be replaced with all Y's non-unit productions (i.e. bb and bXb).

The new CFG, without unit productions, is:
S → aX | a | bab | bb | bXb
X → bb | bXb

(The Y productions are no longer necessary in the CFG).

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

S → AX | a | BAB | BB | BXB
X → BB | BXB
A → a
B → b

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

S → AX | BQ | BB | BR | a
X → BB | BR
Q → AB
R → XB
A → a
B → b

Example8
Convert the following CFG to Chomsky Normal Form (CNF):
S → aY | Ybb | Y
X → /\ | a
Y → aXY | bb | XXa

Solution 8
Step 1 - Kill all /\ productions:

By inspection, the only nullable nonterminal is X.
Delete all /\ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without /\ productions, is:
S → aY | Ybb | Y
X → a
Y → aXY | aY | bb | XXa | Xa | a

Step 2 - Kill all unit productions:

The only unit production is S → Y, where the Y can be replaced with all Y's non-unit productions (i.e. all of them).

The new CFG, without unit productions, is:

S → aY | Ybb | aXY | aY | bb | XXa | Xa | a
X → a
Y → aXY | aY | bb | XXa | Xa | a

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

S → AY | YBB | AXY | AY | BB | XXA | XA | a
X → a
Y → AXY | AY | BB | XXA | XA | a
A → a
B → b

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

S → AY | YR | AQ | AY | BB | XP | XA | a
X → a
Y → AQ | AY | BB | XP | XA | a
A → a
B → b
R → BB
Q → XY
P → XA